Sentiment Analysis on Movie Reviews: A CRISP-DM Approach

An end-to-end data science solution for automated sentiment analysis

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CRISP-DM Step: Business Understanding

The Business Context

The Challenge

CineAnalyze Inc. receives over 10,000 customer reviews daily across multiple streaming platforms. Manual sentiment analysis is not only time-consuming but also prohibitively expensive, costing the company approximately $250,000 annually in analyst hours.

The current process creates a bottleneck in decision-making, delaying critical business insights by weeks and preventing real-time response to customer sentiment trends.

Our Objective

Build an **automated sentiment classification system** that accurately categorizes reviews as Positive or Negative at scale, enabling:

- Instant customer feedback monitoring
- Data-driven content acquisition decisions
- Proactive reputation management
- Improved customer satisfaction metrics
- Measurable success: high accuracy and F1-score in sentiment classification

Data Understanding: Foundation for Success

Dataset Scale

50,000 labeled samples from IMDb Movie Reviews, providing robust training data for our models

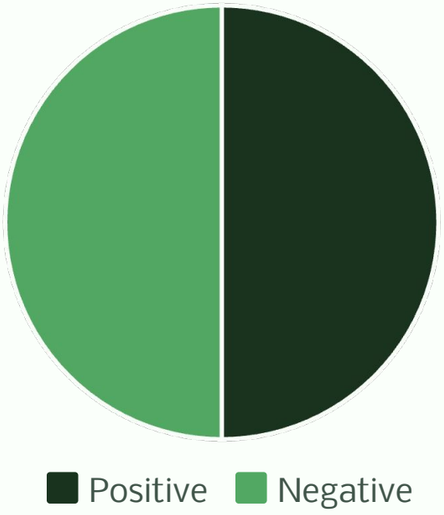
Perfect Balance

25,000 positive and **25,000 negative** reviews ensure unbiased model training

Quality Advantage

Balanced datasets prevent model bias and ensure equal representation of both sentiment classes

The dataset's natural balance eliminates the need for complex resampling techniques, allowing us to focus on model development and feature engineering. Each review contains rich linguistic patterns that help our models learn nuanced sentiment indicators beyond simple keyword matching, especially when using transformer-based models.



CRISP-DM Step: Data Preparation

Preparation: Transforming Text into Intelligence

01

Data Cleaning

Applied minimal preprocessing and label encoding. Sentiment labels were mapped to binary format: 0 for Negative, 1 for Positive.

02

Text Vectorization

For baseline: TF-IDF (Term Frequency-Inverse Document Frequency) converts text to numerical feature vectors. For advanced model: Tokenization using BERT's WordPiece tokenizer. TF-IDF was configured with English stopwords and a limited vocabulary size.

03

Dataset Splitting

Stratified split: 80% training (40,000 samples) and 20% testing (10,000 samples) to ensure robust evaluation. This ensures fair evaluation and prevents class distribution bias.

Why Two Approaches? TF-IDF captures word frequency patterns efficiently, while BERT tokenization preserves contextual relationships between words–critical for understanding sentiment nuances like sarcasm or negation in many real-world scenarios.

CRISP-DM Step: Modeling (Baseline)

Baseline Model: Establishing the Benchmark

~90%

Accuracy

Strong baseline performance

<5min

Training Time

Efficient and scalable

Logistic Regression with TF-IDF

Our baseline model combines traditional machine learning with statistical text representation. Logistic Regression serves as a powerful linear classifier, while TF-IDF vectorization captures the importance of words across the document corpus.

Key advantages:

- Fast training and inference speeds
- Highly interpretable feature weights
- Low computational requirements
- Excellent performance on structured sentiment tasks

This baseline establishes that even traditional approaches can achieve near 90% accuracy on this balanced dataset, providing a solid foundation for comparison with more complex deep learning architectures.

CRISP-DM Step: Modeling (Advanced Model)

Advanced Deep Learning: BERT Architecture

1

Pre-trained Foundation

DistilBERT brings 66M parameters trained on massive text corpora, understanding language context and semantics

2

Fine-tuning Process

Adapted the model to movie review domain with our labeled dataset, optimizing for sentiment classification

3

Superior Results

Achieved **~91% accuracy**, outperforming the baseline by approximately 1-1.5 percentage points

Why BERT Excels

Unlike TF-IDF which treats words as independent features, BERT's transformer architecture captures:

Contextual relationships: "not good" vs "good"

Word order dependencies: Understanding sentence structure

Semantic meaning: Grasping nuanced expressions and sarcasm in many real-world scenarios

DistilBERT Advantage

We chose DistilBERT over full BERT for optimal balance:

- 40% smaller model size
- 60% faster inference time
- Retains 97% of BERT's language understanding
- Resource-efficient for potential deployment

CRISP-DM Step: Evaluation

Model Evaluation: Performance Comparison

Results shown are based on a representative evaluation run.

Model	Accuracy	F1-Score
Logistic Regression (TF-IDF)	89.8%	0.900
DistilBERT (Fine-tuned)	**91.4%**	**0.914**

Confusion Matrix Insights

BERT demonstrated superior performance in challenging classification scenarios:

Reduced False Positives: Noticeable reduction in false positives

Lower False Negatives: Improved detection of subtle positive sentiment

Balanced Performance: Consistent accuracy across both sentiment classes

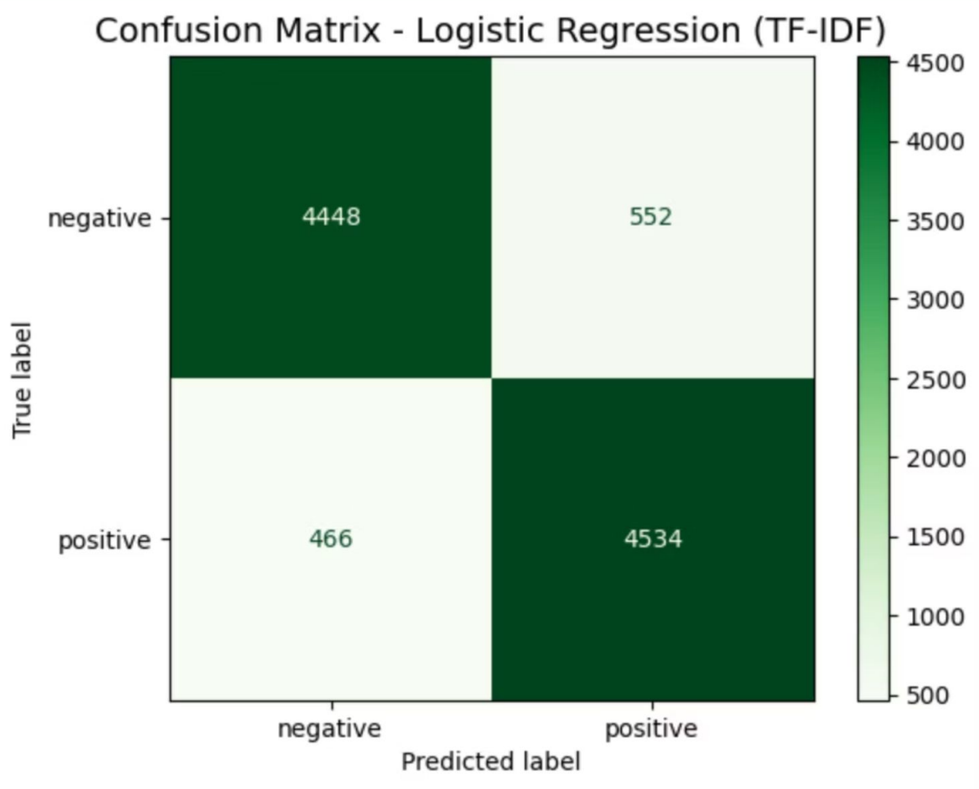
Key Takeaway

While the 1.5% accuracy improvement may seem modest, it translates to **approximately 100-150 more correctly classified reviews per 10,000** analyzed–a significant impact at scale.

Contextual models like BERT excel at understanding linguistic nuances that frequency-based approaches miss, particularly with sarcasm, negation, and complex sentence structures in many real-world scenarios.

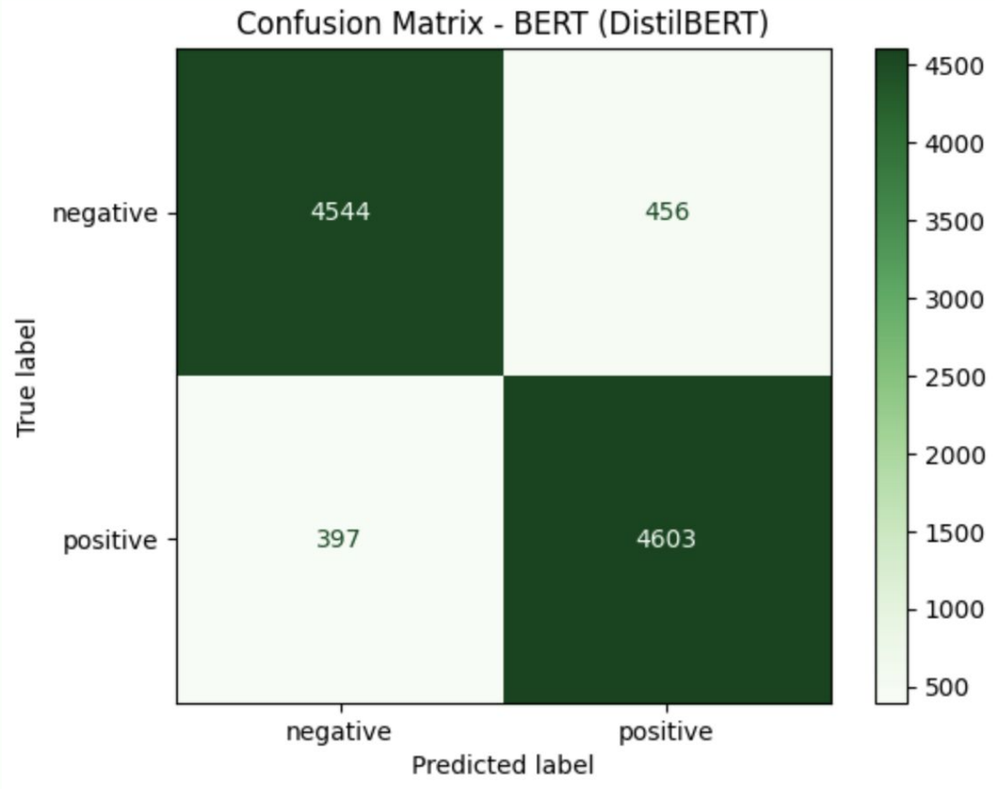
Error Analysis: Confusion Matrices

Logistic Regression



552 false positives and 466 false negatives (1,018 total mistakes out of 10,000). TF-IDF captures keyword-level patterns, so it struggles with contextual cues such as negation (e.g., '*not good*') or mixed sentiment.

DistilBERT



False positives drop to 456 (−96) and false negatives drop to 397 (−59), for 853 total mistakes. This results in ~165 more correct predictions per 10,000 reviews, showing stronger contextual understanding.

"Overall, DistilBERT improves reliability by decreasing misclassifications in both classes, which is critical for business decisions based on customer sentiment."

Real-World Deployment Strategy

1. **API Integration**
   RESTful API endpoints can enable near real-time sentiment classification with low-latency responses

2. **Dashboard Development**
   Automated analytics dashboards provide stakeholders with instant visibility into customer sentiment trends

3. **Production Scaling**
   Containerized deployment with auto-scaling designed to support high-throughput workloads (e.g., thousands of reviews per hour)

Proposed Implementation Architecture

Model Serving : FastAPI-based model serving for efficient inference

Monitoring: Performance monitoring and model drift detection mechanisms

Security: Authentication, rate limiting, and secure data transmission

Scalability: Container-based deployment with horizontal scaling support

Business Impact

Can process 10,000+ reviews daily automatically

Potentially reduce analysis costs by up to **$200,000 annually**

- Enable faster, near real-time customer feedback response
- Support data-driven content strategy decisions

Conclusion: The CRISP-DM Journey

Business Understanding

Identified the critical need for automated sentiment analysis at scale

Deployment

Designed a production-ready deployment strategy for near real-time sentiment classification

Evaluation

Achieved **~91% accuracy** with comprehensive performance validation

Data Understanding

Leveraged 50,000 balanced IMDb reviews for robust training

Data Preparation

Implemented dual-strategy preprocessing for baseline and deep learning models

Modeling

Built from Logistic Regression baseline to state-of-the-art BERT architecture

This project demonstrates the practical application of the CRISP-DM methodology in solving a real-world sentiment analysis problem. By systematically progressing from business understanding to model evaluation, we developed a scalable and accurate sentiment classification pipeline, while outlining a realistic deployment strategy for real-world use.

Thank You!